Probability-Valued Subset Sum: Certified Bounds and Learned Guidance

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Companion note III of III

Abstract. The first note asks when certified coverage can replace explicit search; the second finds no incremental exact subset-sum gain from the tested pairing method. This note changes the question: for fixed weights and a specified random target, what is the probability that some subset reaches it? Witnessed sums and sound exclusions give deterministic bounds on this probability. On 50 held-out 16-item lists, the median lower and upper endpoints from simple rules were 0.8% and 36.0%, while the median exact reachable-target fraction was 24.2%. A learned classifier supplies useful statistical ranking but no per-instance certificate. The bounds and experiments suggest an approximate, costed coverage problem rather than a faster exact decision algorithm.

The probability requires a random experiment

For fixed positive weights $w = (w_1, \ldots, w_n)$ and target t, the decision answer

$$Y(w,t) = \mathbf{1}\left\{ t = \sum_{i=1}^{n} x_i w_i \text{ for some } x \in \{0,1\}^n \right\}$$

is exactly 0 or 1. To obtain a probability, let D be a finite target set, T a random target with distribution μ supported on D, and

$$A(w) = \left\{ \sum_{i=1}^{n} x_i w_i : x \in \{0,1\}^n \right\}.$$

$$q_\mu(w) = \Pr_{T \sim \mu}[Y(w,T) = 1] = \sum_{t \in A(w) \cap D} \mu(t).$$

For a uniform target, $q_\mu(w) = |A(w) \cap D|/|D|$. Thus the new outcome is the chance that a target drawn by a *stated rule* is reachable. Changing D or μ changes the question; the probability is not uncertainty about a fixed target.

Bounds that remain correct for one list

Let $L \subseteq A(w) \cap D$ be distinct sums with explicit subset witnesses. For a positive modulus m, let $R_m(w) = \{a \bmod m : a \in A(w)\}$. Given a finite collection M of moduli, define

$$U = D \cap \bigcap_{m \in M} \{t : t \bmod m \in R_m(w)\}.$$

Proposition 1 (Certified probability interval). For every w, D, μ, L, M defined above,

$$\mu(L) \le q_\mu(w) \le \mu(U), \qquad \mu(B) := \sum_{t \in B} \mu(t).$$

Proof. Every witnessed sum belongs to $A(w)$, giving $L \subseteq A(w) \cap D$. Every attainable sum has an attainable residue modulo every $m \in M$, giving $A(w) \cap D \subseteq U$. Monotonicity of probability mass proves both inequalities. □

Repeated values give another cheap bound. If distinct weight value v_j occurs c_j times, at most $c_j + 1$ copies of that value can be chosen. Hence

$$|A(w)| \le K(w) := \prod_j (c_j + 1).$$

For a uniform target on D, combine the two upper bounds:

$$\frac{|L|}{|D|} \le q_\mu(w) \le \min\left\{\frac{|U|}{|D|}, \frac{K(w)}{|D|}, 1\right\}.$$

The count $K(w)$ is only an upper bound: different count vectors may have the same sum. Exact paired summaries can enlarge L, but their distinct union must be counted, and their construction cost must be charged. More sound exclusions can shrink U.

The interval certifies an additive-ε answer whenever its width is at most ε. Exact q_μ requires zero width. This is the probability analogue of the first note's costed coverage idea: bound quality matters together with the cost of producing it.

A direct check of bound quality

We reused the held-out $n = 16$ lists from the classifier experiment in note II. For each of 50 lists, the target set D was the integer middle 40% of the list's total sum, restricted to targets that passed the experiment's residue tests modulo $8, 10, 16, 32$. Conditional on that filter, targets were drawn uniformly from D. The same residue tests therefore supply no further upper-bound reduction on this particular D.

The lower bound used distinct sums from 512 randomly sampled subsets; the upper bound used $K(w)/|D|$. We enumerated all attainable sums to measure the exact $q_\mu(w)$ *for evaluation only.* Each computed interval contained the exact value.

Table 1: Medians across held-out $n = 16$ lists. Columns are separate medians, not one interval for a representative list.

Family	Lists	Witness lower	Exact q_μ	Count upper
Independent weights	26	0.3%	25.8%	36.9%
Repeated pairs	24	2.2%	24.2%	36.0%
All lists	50	0.8%	24.2%	36.0%

These bounds do not claim a runtime improvement. They show that counting possible selections can sometimes limit the probability without deciding each target, while a small witness budget leaves much of the mass unresolved.

Statistical probabilities and learned guidance

The classifier in note II outputs $\widehat{p}(w, t)$. Since $Y(w, t)$ is fixed once the full instance is known, $\widehat{p}$ is best understood as an estimate of $\Pr(Y = 1 \mid \phi(w, t))$ under a specified input distribution, where ϕ is the model's lossy feature vector. It is distinct from the exact $q_\mu(w)$ for one fixed list.

The held-out experiment drew four targets per independent list. Its 229 YES answers among 720 targets give an observed rate of 31.8% across the 180 held-out lists. A 95% bootstrap interval resampling whole lists is 26.9%–36.7%. Treating each list's four-target average as one bounded independent observation gives a wider Hoeffding 95% interval of 21.7%–41.9% for the mean success rate across that test schedule. On 240 unseen $n = 24$ lists, the observed rate was

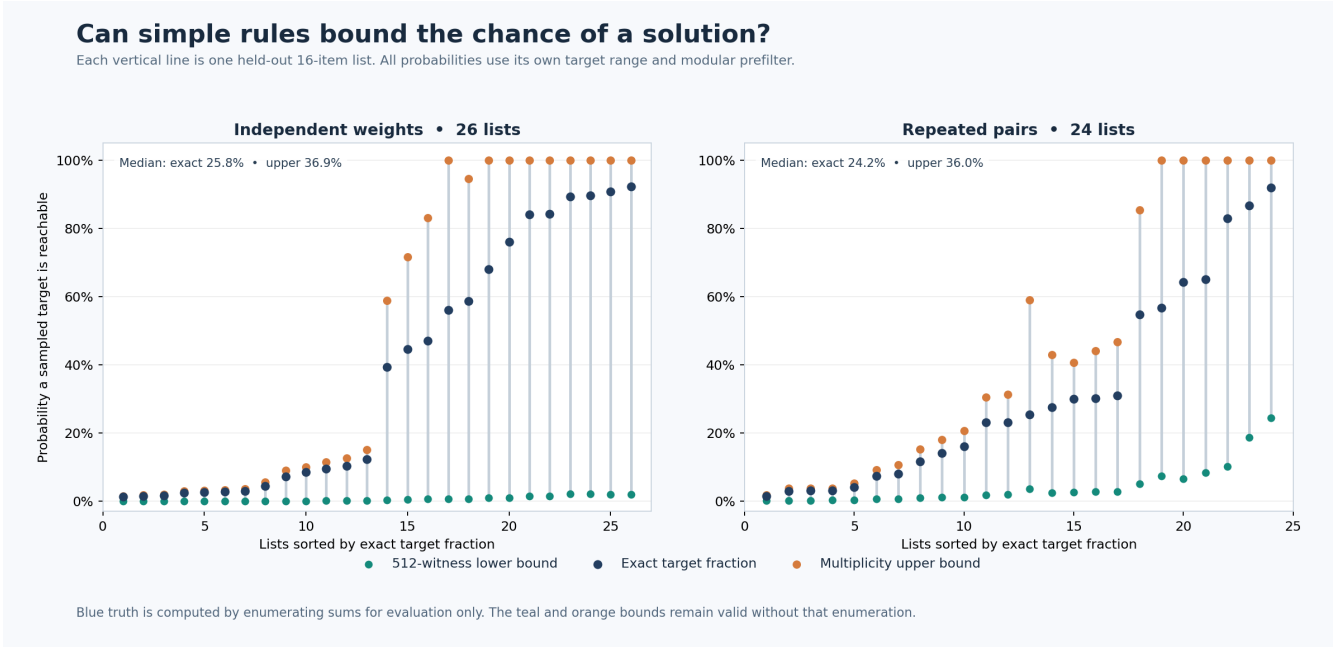


Figure 1: Each vertical segment is a certified interval for one list; the dark point is its exact reachable-target fraction. The low-budget witness bound is often loose.

25.0% across 960 targets; the list-bootstrap interval was 20.9%–29.2%. These rates depend on the synthetic list and target generators.

For a *fixed* list, m independent random targets with exact labels also give the standard sampling bound

$$\Pr\{|\widehat{q} - q_\mu(w)| > \varepsilon\} \le 2e^{-2m\varepsilon^2}.$$

The exact-label cost remains. Sampling uncertainty intervals are different from Proposition 1's deterministic bounds.

At the classifier threshold $\widehat{p} \le 1\%$, 2 of 43 held-out proposed NO answers were actually YES. The list-bootstrap interval for that conditional error fraction was 0%–10.7%; its zero lower endpoint does not certify zero future risk. A learned score can order which witnesses or certificates to seek, but cannot itself certify a NO decision.

Next question

The useful research target is a cheap sequence of witnessed sets L and sound supersets U whose mass gap closes quickly for a meaningful target distribution. One can measure bound width after a fixed budget, or total work to reach a chosen ε, against direct exact enumeration and sampling with exact labels. The present results motivate this experiment; they do not show a new complexity bound or exact subset-sum speedup.

Series and reproducibility. Note I: *Certified Coverage of Exponential Search Spaces.* Note II: *Subset Sum as a Stress Test for Certified Coverage.* Scripts, figures, and held-out predictions: `https://github.com/jonland82/exponential-search-spaces`.